

INTRODUCTION TO DATA SCIENCE AND PROGRAMMING

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SELECTION

WHAT IS SELECTION?

- In programming, **selection** refers to a computer's process of **making decisions** based on rules, also known as **conditions**
- Selection allows programs to make decisions and **respond to different inputs**
- For example, when it's below 10°C, a computer might tell you "It is cold", or if it's above 30°C, it might tell you "It is hot"

<	less than
<=	less than or equal to
>	greater than
>=	greater than or equal to
!=	not equal to
==	is equal to

BOOLEANS

- **Booleans are always either true or false**
- Booleans can be computed using **relational operators**

```
myAge = int(input("What is your age "))  
canDrive = (myAge >= 18)  
isTeenager = (myAge >= 13)
```

BOOLEANS

IF STATEMENTS

- **If** statements allow the computer to **select** what to do
- We give the **if statement** a boolean, which is the rule the computer must follow
- If the rule is not followed, then the computer does the **else statement**

```
myAge = int(input("What is your age "))  
canDrive = (myAge >= 18)  
  
if canDrive:  
    print("You can drive")  
else:  
    print("You cannot drive")
```

IF STATEMENTS

“AND” AND “OR” OPERATOR

- The **and** operator is used when you want **all conditions to be true** for the entire expression to evaluate as True
 - If any one of the conditions is False, the whole expression becomes False
- The **or** operator is used when you want **at least one condition to be true** for the entire expression to evaluate as True
 - The whole expression becomes False only if all conditions are False


```
age = 25
hasTicket = True

if age >= 18 and hasTicket:
    print("You can enter the event")
else:
    print("You cannot enter")

age = 16
hasPermission = True

if age >= 18 or hasPermission:
    print("You can enter the event")
else:
    print("You cannot enter")
```

IF STATEMENTS

ELIF STATEMENTS

- In Python, **elif** is short for “else if” – it is used in conditional statements when you want to test multiple rules **sequentially**
- After an initial if condition, elif allows you to check additional conditions if the previous ones are not met

```
x = 10

if x > 10:
    print("x is greater than 10")
elif x == 10:
    print("x is equal to 10")
else:
    print("x is less than 10")
```

IF STATEMENTS

PRACTICE

Part I